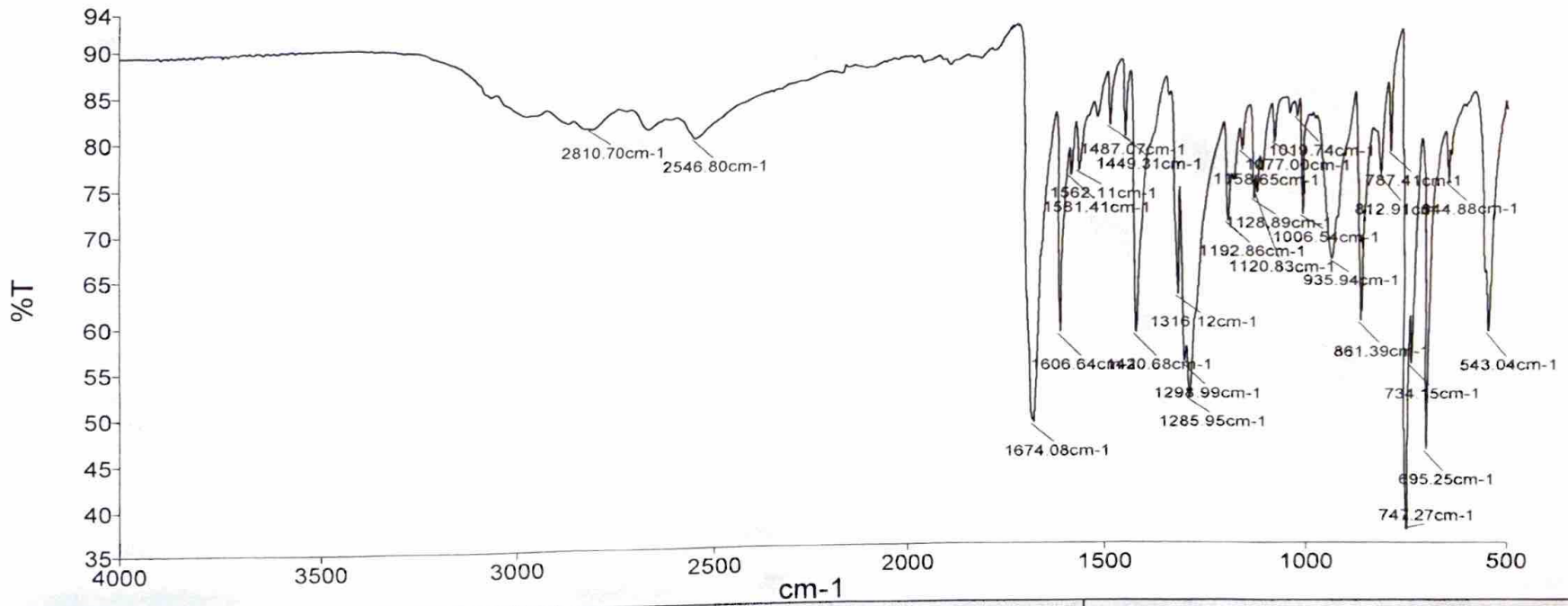
Analyst Administrator

Date Thursday, 1 October, 2026 2:35 PM



Sample Name	Description	Quality Checks
Suzuki_KimDongHwan	Sample 980 By Administrator Date Thursday, October 01 2026	The Quality Checks do not report any warnings for the sample.

DATASHEET — WORKING LOG

LHN

Suzuki–Miyaura Cross-Coupling in Water

Name: KIM DONGHWAN Fumehood: B7 Date: 1 Oct 2026 Photos taken: ☐

1. Masses

4-Bromobenzoic acid (g)	0.3502	Phenylboronic acid (g)	0.2491
n(ArBr) = mass/201.02 (mmol)		Equiv. $PhB(OH)_2$	
Empty plastic bag (g)	0.8342	Bag + product (g)	0.9502
Product (g)		Theoretical (g) = 0.9861 m_{ArBr}	
		Percentage yield (%)	

2. Time log

Pd added (clock)	Out of block	HCl addition done	Into IR lamp

3. Checkpoints

Stage	**Expected** [source]	**Observed / time**	✓
Solids + Na_2CO_3, stirred	all reactants dissolve [M]; clear, colourless [E]		
Pd catalyst solution	pale yellow [M, App. 1]		
After 30 min at 70 °C	*record colour and any solid or grey tint (Pd black)* [E]	no Pd black solids	
HCl addition	gas forms [M] (CO_2 [C]); solid precipitates [E]		
pH paper after HCl	acidic [M]; calc. pH ≈ 0.7 [C]	0 ~ 1 ?	
Hot ethanol, solution 1	all material dissolves with ~~minimal ethanol~~ [M]	boiled longer ~> add slightly more EtOH	
Cooling to RT, then ice	crystals form (recrystallisation) [M]		
Dried product	white solid [E]	reason of small yield?	

Sources: [M] stated in the lab manual; [C] calculated from manual quantities; [E] general expectation, not stated in the manual. The Expected column is a **checkpoint, not an answer**: record what you actually see; a mismatch is a result worth discussing.

4. Ethanol used in recrystallization & catalyst metrics

Hot EtOH added from sol. 2 (mL, est.)		n(product) (mmol)	
TON = $n_{product}/(2.5 \times 10^{-4}$ mmol)		TOF = TON/30 min (1/min)	
IR file name		NMR tube no. / solvent	

5. Other observations

To be completed at the end of the session		
Student signature	GTA name	GTA signature

This is a working log. The **official** datasheet (lab manual p. 7) must still be printed, signed by the GTA and submitted with the report.